ARYAMAN RAO

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EDUCATION

University of Michigan — **Ann Arbor, MI**
Master of Science: Electrical and Computer Engineering (Track: Robotics); GPA: 4.0/4.0 — *Expected May 2026*
Coursework: Foundations of Computer Vision, Advanced Computer Vision, Self-Driving Cars, Algorithmic Robotics, Mobile Robotics, Mathematics for Robotics, Programming for Robotics, PJTL Techlabs at MCity, Directed Study: Reinforcement Learning.

Delhi Technological University (DTU) — **New Delhi, India**
Bachelor of Technology: Electrical Engineering (Minor: Machine Learning); CGPA: 8.66/10 (3.8/4) — *May 2024*

WORK EXPERIENCE

MCity – University of Michigan Transportation Research Institute — **Ann Arbor, MI**
Software Engineering Intern — *Jun 2025 – Present*

- Developed C++ ROS2 localization and trajectory (lateral/longitudinal) planning modules for a **semi co-driving automotive** system in Autoware, enabling closed-loop control with validated simulation (VI-grade and CARLA); collaborated with Stellantis engineers.
- Built a **synthetic data generation** pipeline via Parallel Domain, using lane-segmented HD maps to create diverse simulation scenarios and produce map-based multi-modal (RGB, LiDAR, Radar) datasets, helping reduce the sim-to-real gap.
- Designed and implemented sensor calibration test rigs for a 6-camera, 4-LiDAR setup and performed **multi-sensor calibration and validation** to achieve high-accuracy fusion and localization using open-source calibration tools and MATLAB.

Buyutech (Techlabs at MCity**)** — **Ann Arbor, MI**
ADAS Engineer — *Aug 2024 – Dec 2024*

- Assembled and configured a stereo-camera calibrated setup with Stereolabs ZEDX cameras and Jetson Orin for a real-time depth estimation; performed **hardware bring-up, debugging, and validation** of multi-sensor synchronization on-vehicle at MCity.
- Achieved a 22ms inference time with 1.57% error on an optimized stereo-matching model using AWS EC2 Linux.

PROJECT EXPERIENCE

Efficient Vision-Language-Action Models for Autonomous Driving | Self-Driving Cars — **Ann Arbor, MI**

- Integrated training-free Token Merging (ToMe) technique into state-of-the-art VLA models like SimLingo and ReCogDrive, achieving up to **60% reduction** in inference GFLOPs with significant latency and memory savings.
- Evaluated on Bench2Drive and NAVSIM in closed-loop testing using 4 GPUs with synchronized parallel evaluation, showing an average ~10% relative PDMS drop and quantifying the efficiency–safety tradeoff in VLA driving models.

End-to-End Vision-Language Driving Planning Model | Waymo Dataset (ongoing) — **Ann Arbor, MI**

- Developed a 4-D transformer-based planner using 360° images and kinematic states from 1TB+ of real-world driving data.
- Designed a multimodal planning pipeline using CLIP as the backbone, extracting and fusing representations (Depth, Temporal, BEV) for scene understanding, and using Qwen for LLM-based reasoning to predict 5-s future agent trajectories autoregressively.
- Optimized architecture using **CUDA/TensorRT optimization** on Vision Language Model to speed up training on SLURM cluster.

Motion Planning Performance Analysis for Autonomous Robotics | Algorithmic Robots — **Ann Arbor, MI**

- Empirically evaluated sampling-based motion planners including RRT-connect, RRT*, Informed RRT*, BiRRT*, and PRM* in obstacle-dense environments on a 6-DOF robotic arm, **analyzing trade-offs** in planning time, path optimality, and smoothness.
- Built a unified benchmarking pipeline using joint-space and task-space metrics, with shortcut-based trajectory smoothing.

SuPerMVO: Monocular Visual Odometry | Mobile Robotics — **Ann Arbor, MI**

- Built a monocular VO pipeline (SuPerMVO) using grey-scale KITTI images with extract keypoints using combined SuperPoint and SuperGlue architectures, estimated 6D pose estimation via OpenCV's built-in functions.
- Integrated GTSAM for loop closure, **improving pose accuracy and computation by 6–7%** over benchmark ORB-based pipelines.

Learning-based Planners (Reinforcement Learning) | Michigan Robotics Department — **Ann Arbor, MI**

- Programmed hierarchical planners by integrating Reinforcement Learning (TD3, SAC) and MPC for Google DeepMind Control tasks in **MuJoCo**; built a Python-based DeepMind Control to Gym wrapper and applied model-based RL concepts.

SKILLS

Languages: Python, C++, C, Bash, MATLAB/Simulink, Julia, R, SQL, JAX, Java.
Frameworks & Tools: PyTorch, TensorFlow, ROS2, OpenCV, CUDA, TensorRT, Docker, Git, RestAPI, AWS, GCP, Kubernetes.
Simulation & Autonomy: Autoware, CARLA, MuJoCo, Isaac Sim, VI-grade, Gazebo, nuPlan, Unreal Engine, DeepMind Control.
Robotics Skills: Motion Planning (Waymo), Localization (Kalman Filtering, SuPerMVO), Sensor Fusion (LiDAR, Camera, IMU, GPS), Vehicle Control (MPC, LQR, RL), Camera–LiDAR Calibration, Visual SLAM (ORB-SLAM), Cloud Deployment.

SELECTED PUBLICATIONS and RESEARCH (Citations: 44+)

- Influence Maximization in Social Networks using discretized Harris' Hawks Optimization: **Applied Soft Computing Journal**
- DQSSA: A Quantum-Inspired Solution for Maximizing Influence in Online Social Networks **(AAAI '24)**
- COPD-FlowNet: Elevating Non-invasive COPD Diagnosis with CFD Simulations (Student Abstract) **(AAAI '24)**